1. $\int \left(12x^2 - 4x + 1\right) dx = \ldots$

Jawab:

Pakai aturan pangkat: $\int x^n \, dx = \dfrac{x^{n+1}}{n+1} + C$

$$\int 12x^2 \, dx = 12 \cdot \frac{x^3}{3} = 4x^3$$

$$\int -4x \, dx = -4 \cdot \frac{x^2}{2} = -2x^2$$

$$\int 1 \, dx = x$$

$$\int \left(12x^2 - 4x + 1\right) dx = 4x^3 - 2x^2 + x + C$$

Cek (turunkan balik):

$$\frac{d}{dx}\left(4x^3 - 2x^2 + x + C\right) = 12x^2 - 4x + 1 \quad \checkmark$$

$\boxed{4x^3 - 2x^2 + x + C}$

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2. $\lim\limits_{x \to 0} \dfrac{e^x - 1 - x}{x^2} = \ldots$

Jawab:

Substitusi $x = 0$:

$$\frac{e^0 - 1 - 0}{0^2} = \frac{0}{0} \quad \text{(tak tentu} \rightarrow \text{L'Hôpital)}$$

$$\begin{aligned}
\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} &= \lim_{x \to 0} \frac{e^x - 1}{2x} && \text{(masih } \tfrac{0}{0}\text{, L'Hôpital lagi)} \\
&= \lim_{x \to 0} \frac{e^x}{2} \\
&= \frac{e^0}{2} \\
&= \frac{1}{2}
\end{aligned}$$

$\boxed{\dfrac{1}{2}}$

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3. $\lim_{x \to 0^+} x^x = \ldots$

Jawab:
Bentuknya 0^0 (tak tentu).
Misal $y = x^x$, maka

$$\ln y = \ln x^x = x \ln x \quad \text{(bentuk } 0 \cdot (-\infty))$$

Ubah jadi pecahan:

$$\begin{aligned}
\lim_{x \to 0^+} \ln y &= \lim_{x \to 0^+} \frac{\ln x}{\frac{1}{x}} && \text{(bentuk } \tfrac{-\infty}{\infty} \to \text{L'Hôpital)} \\
&= \lim_{x \to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} \\
&= \lim_{x \to 0^+} \frac{1}{x} \cdot \left(-x^2\right) \\
&= \lim_{x \to 0^+} (-x) \\
&= 0
\end{aligned}$$

Karena $\ln y \to 0$:

$$y \to e^0 = 1$$

$\boxed{1}$

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4. Tentukan konvergen/divergen: $\displaystyle\int_1^\infty \frac{\ln x}{x^2}\, dx$

Jawab:
Batas atas $\infty \to$ integral tak wajar, ubah ke limit:

$$\int_1^\infty \frac{\ln x}{x^2}\, dx = \lim_{b \to \infty} \int_1^b \frac{\ln x}{x^2}\, dx$$

Integral parsial $\displaystyle\int u\, dv = uv - \int v\, du$

$$\begin{aligned}
u &= \ln x & dv &= x^{-2}\, dx \\
du &= \frac{1}{x}\, dx & v &= -\frac{1}{x}
\end{aligned}$$

$$\int \frac{\ln x}{x^2}\,dx = \ln x \cdot \left(-\frac{1}{x}\right) - \int \left(-\frac{1}{x}\right)\frac{1}{x}\,dx$$
$$= -\frac{\ln x}{x} + \int \frac{1}{x^2}\,dx$$
$$= -\frac{\ln x}{x} - \frac{1}{x}$$

Masukkan batas:

$$\int_1^b \frac{\ln x}{x^2}\,dx = \left[-\frac{\ln x}{x} - \frac{1}{x}\right]_1^b$$
$$= \left(-\frac{\ln b}{b} - \frac{1}{b}\right) - \left(-\frac{\ln 1}{1} - 1\right)$$
$$= -\frac{\ln b}{b} - \frac{1}{b} + 1 \qquad (\ln 1 = 0)$$

Cari limitnya, $\dfrac{\ln b}{b}$ bentuk $\dfrac{\infty}{\infty}$ $\rightarrow$ L'Hôpital:

$$\lim_{b\to\infty} \frac{\ln b}{b} = \lim_{b\to\infty} \frac{\frac{1}{b}}{1} = 0$$

$$\lim_{b\to\infty} \left(-\frac{\ln b}{b} - \frac{1}{b} + 1\right) = -0 - 0 + 1$$
$$= 1$$

Limitnya ada (hingga), jadi integralnya **konvergen**.

$$\boxed{\int_1^\infty \frac{\ln x}{x^2}\,dx = 1 \text{ (konvergen)}}$$